Fixing Missing Initial Velocity for Newly Entering Inlet Droplets

Droplet rollout modeling notes

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1 Problem

The canonical droplet dataset stores each droplet state as

$$(x, y, v_x, v_y, c),$$

where c is circularity. Velocities are currently computed by finite differencing each track:

$$v_x(t) = x(t) - x(t-1), \qquad v_y(t) = y(t) - y(t-1).$$

At the first visible frame of a track, there is no previous observation. Therefore v_x and v_y are undefined and become NaN.

During model-window construction, non-finite values in the input history are converted to zero. For newly entering inlet droplets this silently converts an unknown inlet velocity into a stationary state. This is physically wrong: a droplet appearing at the inlet is not stationary; it has just entered the field of view.

2 Observed Impact

The failure case for track 3032 illustrates the problem. At the Markovian input frame, the droplet has just appeared near the upper inlet, but its velocity is missing:

$$y \approx 3.9, \qquad v_x = \mathrm{NaN}, \qquad v_y = \mathrm{NaN}.$$

After NaNs are converted to zero, the model receives an input resembling a stationary droplet near the inlet. The true future motion, however, rapidly reaches approximately 11 pixels per frame in the y direction.

In validation diagnostics, this start-state issue affected 110 evaluable trajectories, or about 0.62% of all evaluable validation trajectories. It was highly concentrated at the inlet: 109 of 110 affected cases had input $y < 10$ pixels. Although rare by sample count, these cases were strongly associated with severe channel-admissibility failures in the geometry-aware Markovian rollout.

3 Scientific Interpretation

This is a data representation issue, not a model architecture issue. The observed state is intended to include velocity. For newly entering droplets, the velocity is not measured at the first visible frame, but the current preprocessing turns that missing value into a literal zero. The model is therefore asked to interpret an ambiguous and physically misleading state.

The issue is especially important for Markovian models, because they receive only the current state. A history-based model can often infer motion from previous frames when they exist, but a one-frame Markovian model cannot infer an inlet velocity from a single position if the velocity channel has been zeroed.

4 Required Correction

The correction should be applied once, when constructing the canonical dataset, before train/validation/test window construction and before any model-specific data loading. This ensures that all models consume the same corrected state representation.

Let $\mathcal{I}$ denote finite velocity observations in the incoming inlet region. The inlet prior velocity is

$$\bar{v}_x^{\text{inlet}} = \frac{1}{|\mathcal{I}|} \sum_{i \in \mathcal{I}} v_{x,i}, \qquad \bar{v}_y^{\text{inlet}} = \frac{1}{|\mathcal{I}|} \sum_{i \in \mathcal{I}} v_{y,i}.$$

For newly visible inlet droplets only, replace missing velocity components:

$$v_x \leftarrow \bar{v}_x^{\text{inlet}} \quad \text{if} \quad v_x = \text{NaN},$$

$$v_y \leftarrow \bar{v}_y^{\text{inlet}} \quad \text{if} \quad v_y = \text{NaN}.$$

The patch must not be applied to arbitrary missing velocities elsewhere in the channel. The intended criterion is:

1. the row is the first visible frame of its track,
2. the droplet is in the incoming inlet region,
3. at least one of v_x, v_y is non-finite.

5 Diagnostics

The canonical dataset build should report:

- the inlet-region criterion,
- $\bar{v}_x^{\text{inlet}}$,
- $\bar{v}_y^{\text{inlet}}$,
- number of droplet rows patched,
- number of velocity values patched,
- number of affected unique tracks.

6 Expected Result

After rebuilding the canonical dataset, first-visible inlet droplets should no longer enter model inputs with zeroed velocities. History-20, Markovian, and geometry-aware Markovian models should all train and evaluate from the same corrected canonical state tensor. No model architecture, rollout logic, boundary injection, loss function, geometry mask, or ellipse construction should change.